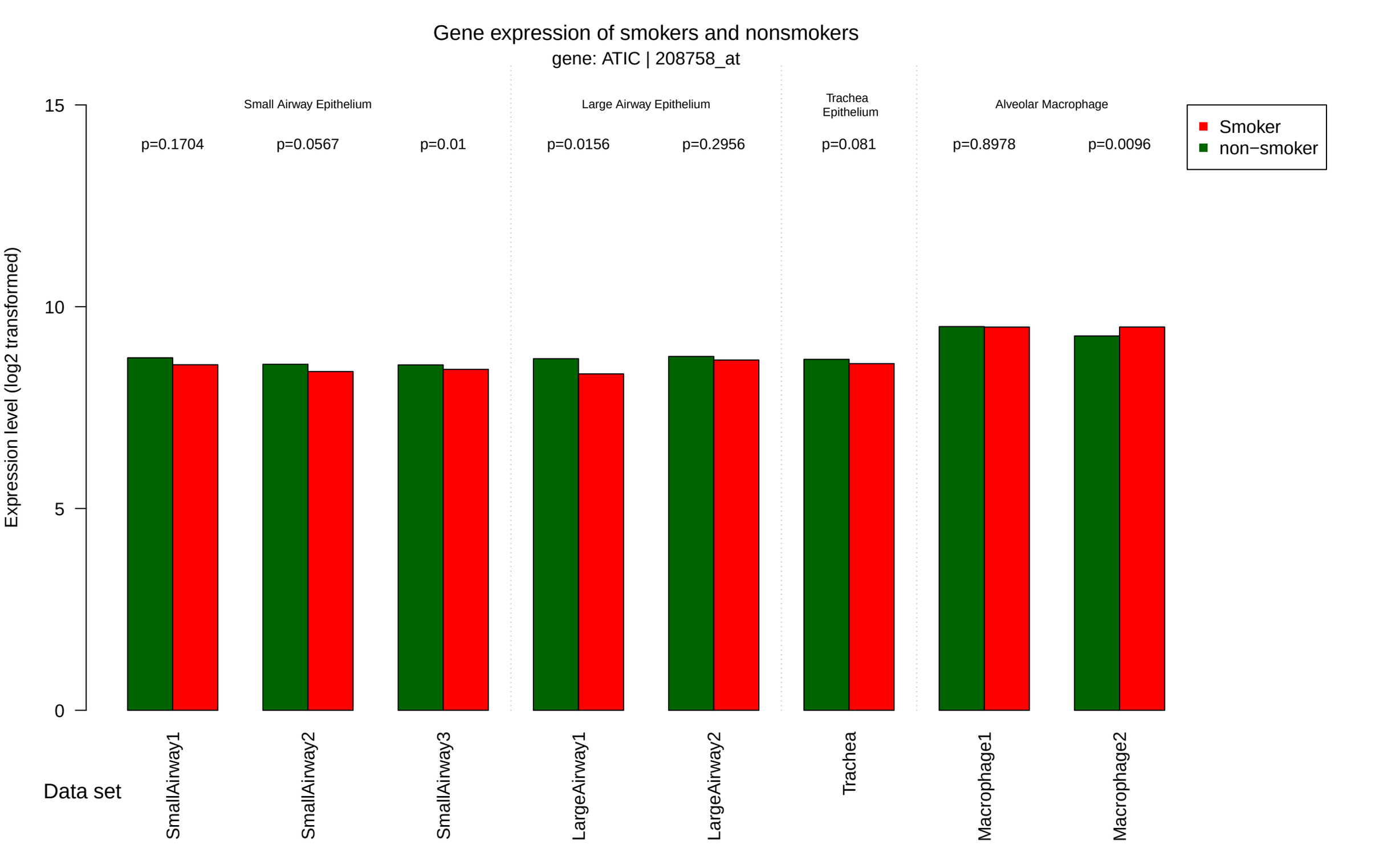
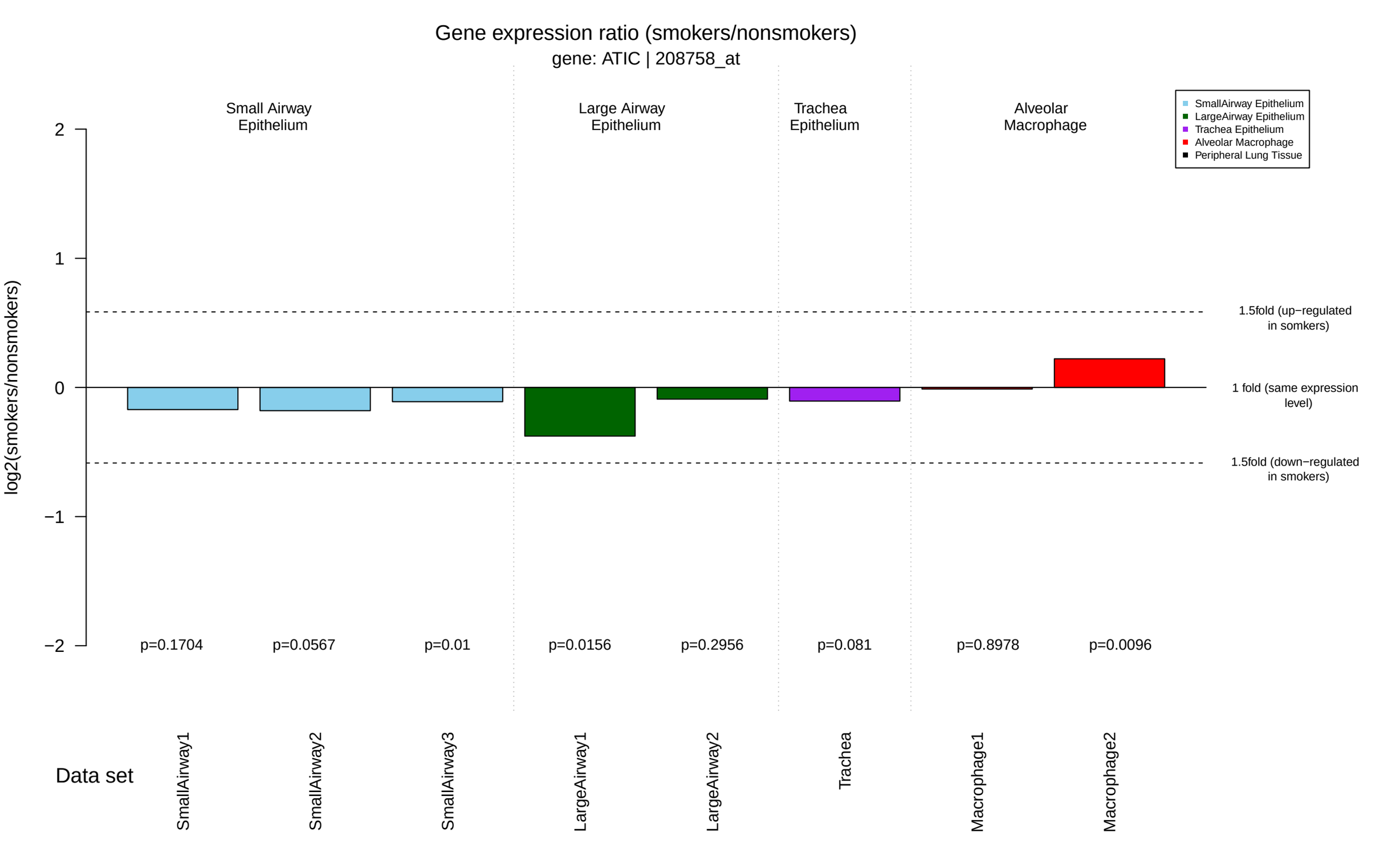
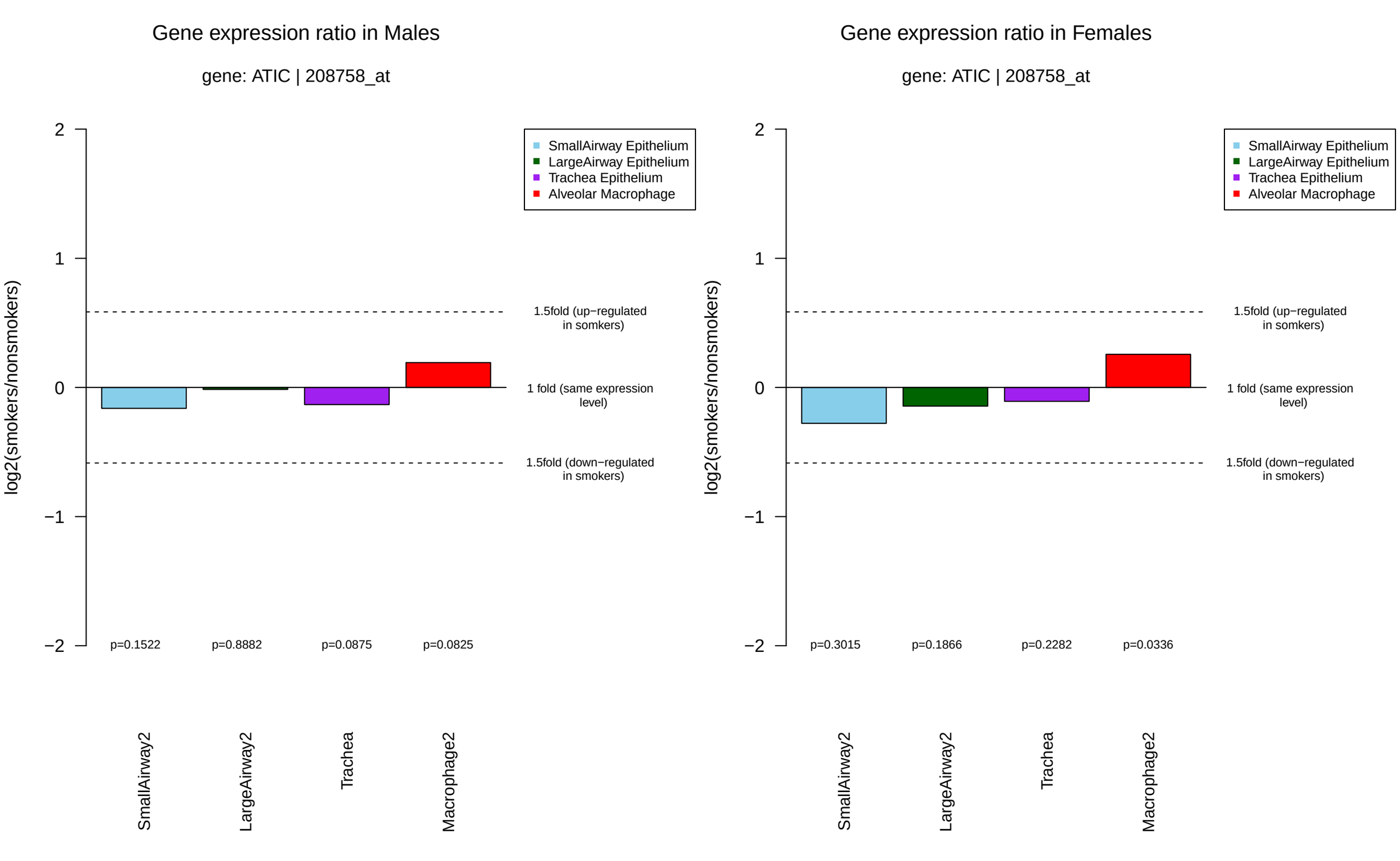
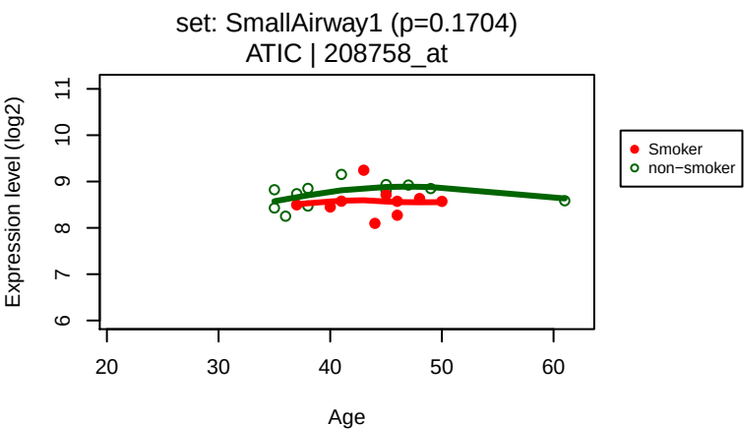
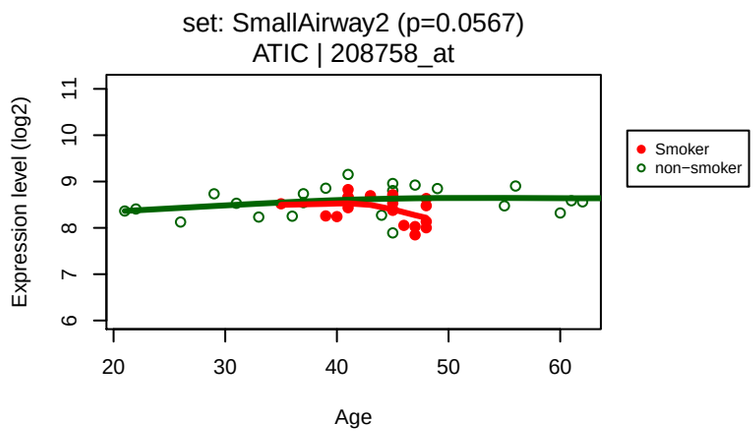
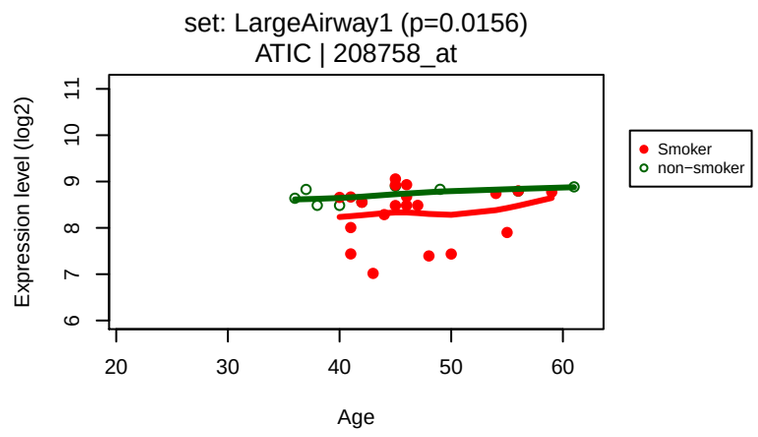

Gene expression of smokers and nonsmokers
gene: ATIC | 208758_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1704
p=0.0567
p=0.01
p=0.0156
p=0.2956
p=0.081
p=0.8978
p=0.0096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

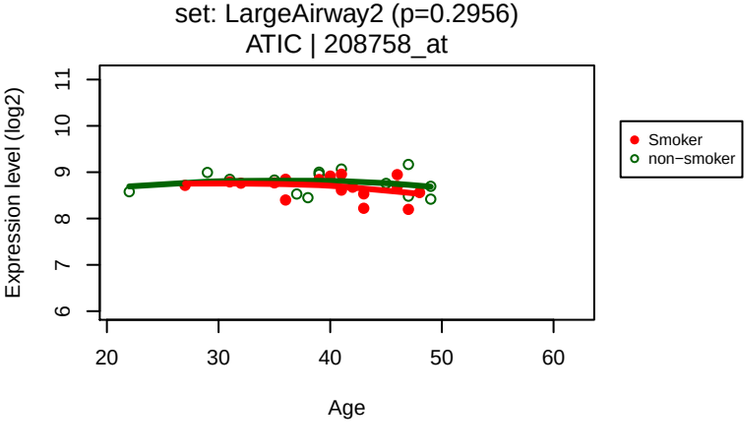

Gene expression ratio (smokers/nonsmokers)
gene: ATIC | 208758_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1704
p=0.0567
p=0.01
p=0.0156
p=0.2956
p=0.081
p=0.8978
p=0.0096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

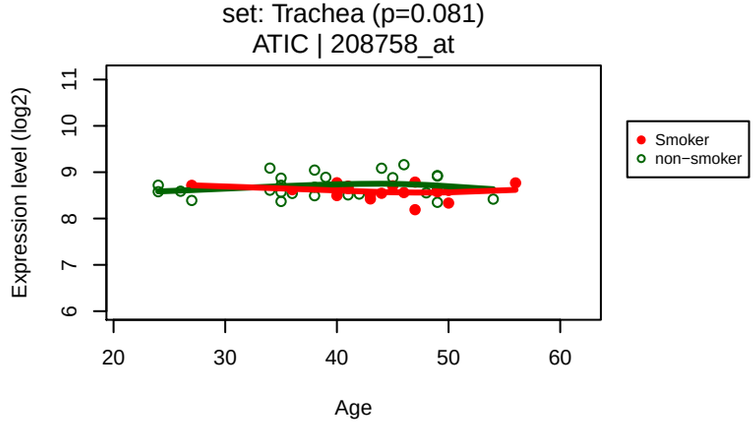

Gene expression ratio in Males
gene: ATIC | 208758_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1522
p=0.8882
p=0.0875
p=0.0825
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATIC | 208758_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3015
p=0.1866
p=0.2282
p=0.0336
SmallAirway2
LargeAirway2
Trachea
Macrophage2

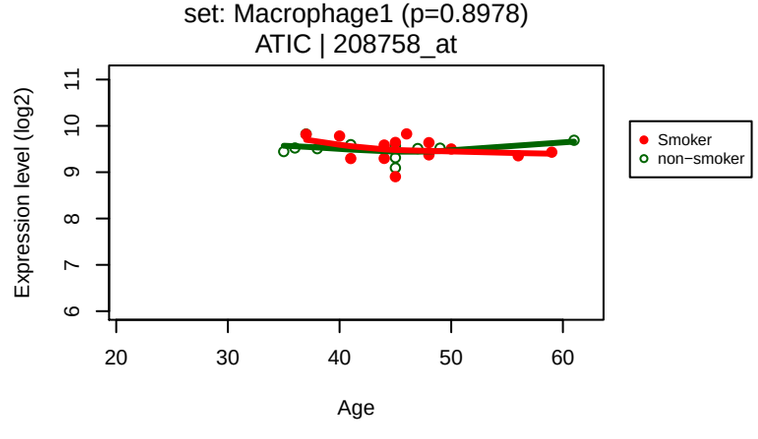

set: SmallAirway1 (p=0.1704)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker

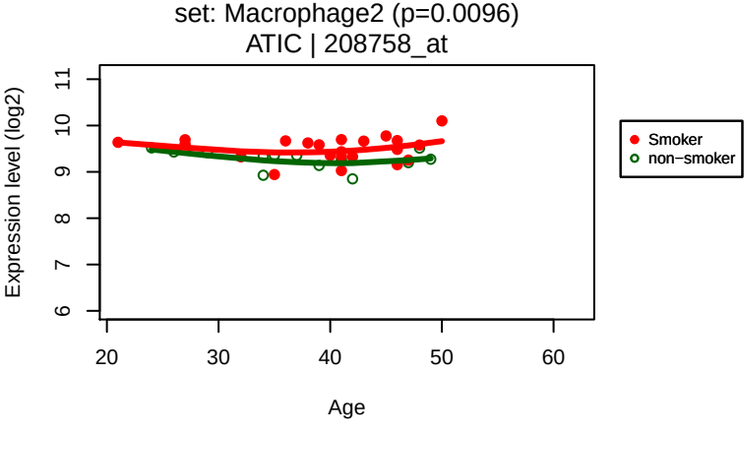

set: SmallAirway2 (p=0.0567)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker

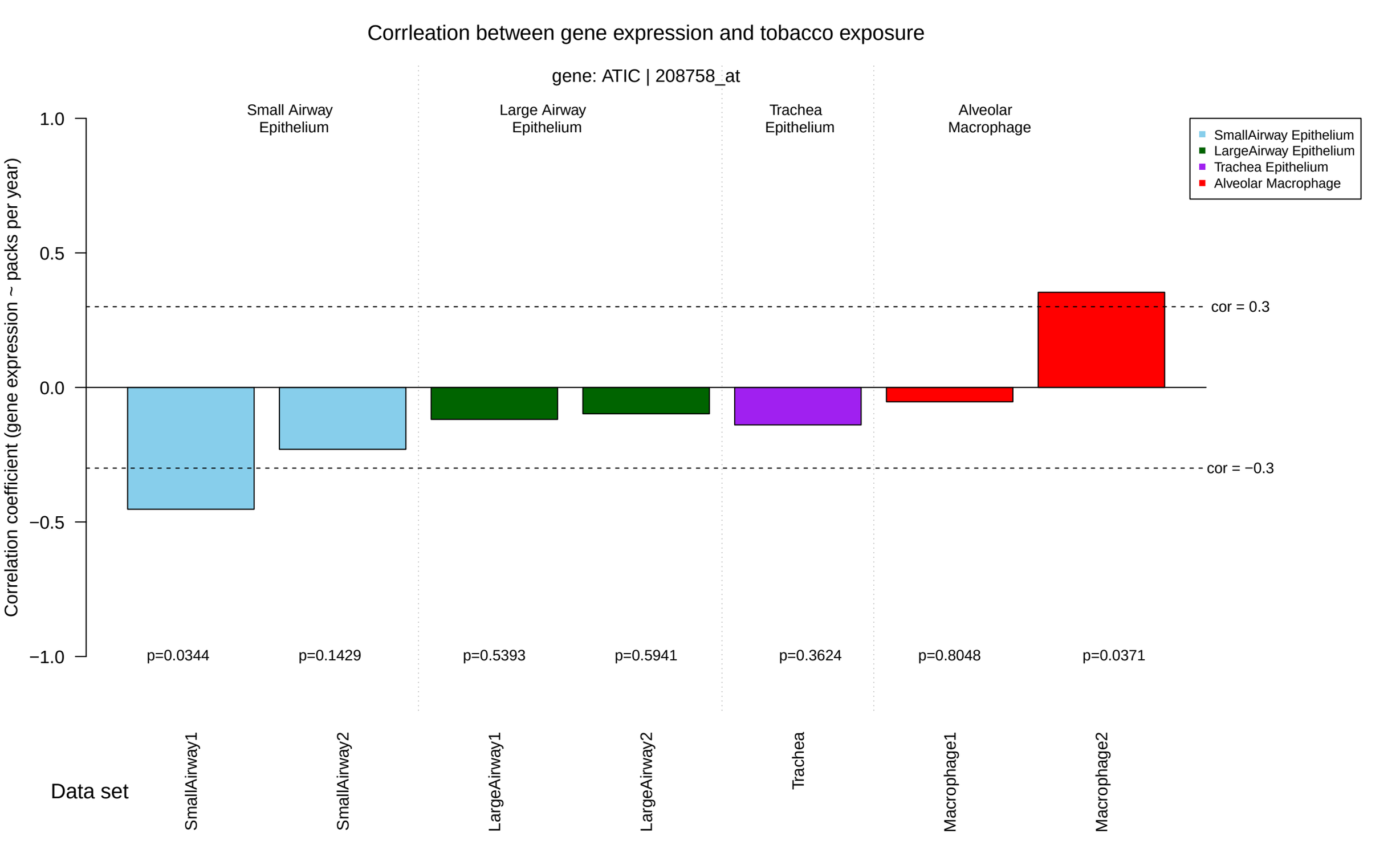

set: LargeAirway1 (p=0.0156)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2956)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.081)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8978)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0096)
ATIC | 208758_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATIC | 208758_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0344
p=0.1429
p=0.5393
p=0.5941
p=0.3624
p=0.8048
p=0.0371
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2